

VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION–LANGUAGE MODELS

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ABSTRACT

Vision–Language Models (VLMs) are compelling zero-shot goal-conditioned value functions, yet frozen pre-trained representations leave important generalization and temporal reasoning capacity unused. We introduce VITA, a zero-shot value function learning method that directly strengthens both capabilities through test-time adaptation. At inference, VITA updates a lightweight adaptation module with a gradient step on a meta-learned self-supervised loss, so each test-time update is trained to improve value estimation. Sequential updates over a trajectory turn the module parameters into temporal memory, directly addressing the temporal reasoning gap of frozen VLM value estimators. To prevent shortcut learning, we introduce a dissimilarity-based sampling strategy that deliberately selects semantically diverse trajectory segments during training. On real-world robotic manipulation tasks, VITA generalizes from a single training environment to out-of-distribution tasks, environments, and embodiments, and outperforms the state-of-the-art zero-shot autoregressive VLM method. We further show that VITA provides effective zero-shot reward shaping for offline reinforcement learning: on Meta-World, the resulting multi-task policies exceed policies trained with the simulator’s fuzzy-logic dense rewards.

1 INTRODUCTION

Vision-Language Models (VLMs) have established a powerful recipe for broad generalization: learn from large-scale, unstructured web data without human supervision (Radford et al., 2021; Alayrac et al., 2022). Generalist policies for robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024) have advanced rapidly, but they still do not match this breadth because they depend heavily on expert demonstrations (Ho & Ermon, 2016). World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) offer a decisive path beyond that bottleneck by letting agents learn inside simulated environments, with impact across robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). As world models become capable of generating realistic visual trajectories in diverse, open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025), the central obstacle is no longer data generation alone but supervision: *how can agents learn from videos at scale without relying on human supervision?*

One direct strategy is to learn policies from expert visual trajectories by inferring actions from visual transitions (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a). Although these latent actions can in principle be mapped to executable controls at deployment, doing so remains difficult in practice. A more broadly reusable route is to learn a universal *goal-conditioned value function* that serves directly as a zero-shot reward-shaping signal for RL and as supervision for imitation learning (Chen et al., 2021; Ma et al., 2022). Under this formulation, value estimation becomes the problem of predicting how far an agent has progressed toward a language-specified goal from visual observations and a natural language task description (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025).

Existing VLM value estimators expose a clear gap. Contrastive VLM methods (Radford et al., 2021) use similarity between task descriptions and visual observations in a shared multimodal space to ground zero-shot reward shaping in semantic context (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024), but they do not model the temporal context required to distinguish visually

similar states that occur at different task stages, such as folding versus unfolding a shirt. Autoregressive VLMs (Alayrac et al., 2022) add trajectory context by conditioning on full visual histories in the prompt, but they inherit a monotonic prediction bias from chronologically ordered pre-training data (Ma et al., 2024). Thus, both architectures rely on fixed pre-trained representations (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023) in a way that constrains zero-shot generalization and temporal reasoning (Pătrăucean et al., 2023). Recent alternatives pursue large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), or demo-based reward specification (Sontakke et al., 2023); our goal is to obtain the missing adaptation capability without requiring those additional resources.

We introduce VITA, a zero-shot value function learning method that makes test-time adaptation the mechanism for both generalization and temporal reasoning. On real-world robotic manipulation tasks, VITA outperforms the state-of-the-art zero-shot approach (Ma et al., 2024). Rather than relying on additional large-scale pre-training or task-specific demonstrations, VITA adapts online to the semantic and temporal context of each trajectory. At inference, a lightweight adaptation module is updated in negligible time by a gradient step on a meta-learned self-supervised loss (Sun et al., 2024), with the update explicitly trained to improve value estimation (Finn et al., 2017). Sequential updates over the test trajectory encode history into the estimator parameters (Sun et al., 2024), directly targeting the temporal reasoning failure mode of prior VLM value functions. We further introduce dissimilarity-based sampling to counter shortcut learning (Geirhos et al., 2020) by forcing training batches to emphasize semantic diversity. The resulting method succeeds across the core requirements for a useful value function: distribution-shift generalization, expert-versus-non-expert discrimination, and zero-shot reward shaping for offline RL (Levine et al., 2020).

Our contributions are direct:

- We introduce VITA, a test-time adaptation method that turns contrastive VLM representations into stronger zero-shot value functions with improved generalization and temporal reasoning, without task-specific demonstrations or additional large-scale pre-training.
- We show that VITA generalizes from a single training environment to out-of-distribution robotic manipulation tasks, environments, and embodiments, outperforming the state-of-the-art zero-shot method (Ma et al., 2024).
- We demonstrate that VITA’s zero-shot value estimates are strong reward-shaping signals: on Meta-World MT10, they produce offline RL policies that surpass policies trained with the simulator’s fuzzy-logic dense rewards (Yu et al., 2020).

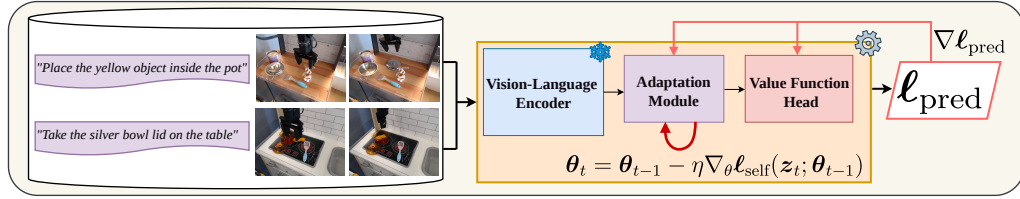
2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

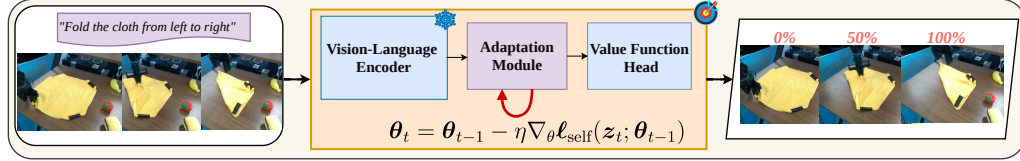
We treat task progress estimation in video trajectories as goal-conditioned value function estimation, with reward grounded in task completion. This framing lets us learn a goal-conditioned value function by predicting completion progress from vision–language representations. Formally, the value function is defined as: $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$ which maps an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to a scalar value indicating predicted progress toward the goal. We set $V(o_t; g) = 0$ at task start and $V(o_t; g) = 1$ at completion. For expert demonstrations, temporal position provides a strong supervision signal because progress is typically monotonic along successful trajectories (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$, we define temporal progress using normalized timestep indices: $V^{\pi_E}(o_t; g) = \frac{t}{T}$ where T is the trajectory length. Thus, temporal progress supplies direct supervision for learning the goal-conditioned value function V .

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) treats inference itself as a self-supervised learning problem (Goodfellow et al., 2016): the model updates parameters on each test instance without labels. Although TTT was introduced for static image classification (Sun et al., 2020), recent sequence-modeling variants (Sun et al., 2024; Wang et al., 2023) show that a parametric adaptation



(a) Training: Meta-learning a goal-conditioned value estimator that is optimized for test-time adaptation



(b) Inference: Producing zero-shot value estimates on OOD trajectories through sequential test-time adaptation

Figure 1: **Overview of VITA.** VITA meta-learns a goal-conditioned value function whose test-time updates deliver zero-shot generalization to out-of-distribution trajectories.

module f_{adapt} can be updated at every timestep by a gradient step on a self-supervised loss. This mechanism gives TTT a distinct temporal-memory structure. Standard sequence models store history in activations, such as hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key-value pairs (Vaswani et al., 2017); TTT instead writes history into the parameters of f_{adapt} through sequential inference-time updates. The parameters of f_{adapt} therefore act as implicit memory that preserves temporal order. Sun et al. (2024) further make the self-supervised task learnable under the gradient-based meta-learning paradigm (Finn et al., 2017): in this setting (Sun et al., 2024), the loss is parameterized by learnable linear projections and optimized for downstream prediction accuracy after the test-time update. We bring this capability to goal-conditioned value function learning, using sequential updates to capture temporal context and meta-learning the self-supervised task specifically to improve value estimation.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

Our goal-conditioned value function estimator has three deliberately separated modules: (1) a multimodal encoder that extracts joint vision-language representations from visual trajectories and task descriptions; (2) a test-time adaptation module updated by a self-supervised loss meta-learned for value estimation; and (3) a regression head that converts adapted representations into value estimates.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We use a frozen contrastive vision-language encoder, CLIP (Radford et al., 2021), as the semantic backbone for visual observations and goal descriptions. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a language task description g , we concatenate the two representations at each timestep t to form a sequence of joint multimodal representations $(z_1, z_2, \dots, z_T)$, where $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$, and ϕ_v and ϕ_g denote the visual and language encoders, respectively. Because CLIP is pre-trained with a contrastive objective that aligns paired image-text inputs in a shared space (Radford et al., 2021), visual observations that are semantically closer to goal completion tend to lie closer to the goal-description representation. VITA builds on this semantic signal and adds the temporal adaptation that frozen similarity alone lacks.

3.1.2 TEST-TIME ADAPTATION

To make multimodal representations respond to the semantic and temporal context of each trajectory, we place an adaptation module f_{adapt} at the center of the estimator, following the test-time training

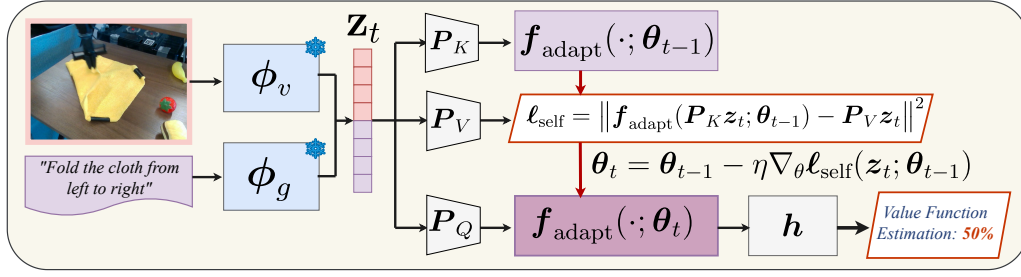


Figure 2: **Test-time Adaptation.** During inference, at each timestep t , VITA updates f_{adapt} with a gradient step on the meta-learned self-supervised loss ℓ_{self} , turning sequential updates into temporal memory.

paradigm in 2.2. At inference, f_{adapt} is updated online by a gradient step on a meta-learned self-supervised loss ℓ_{self} . The loss is a reconstruction objective built from learnable linear projections P_K and P_V , and these projections are meta-learned so the resulting test-time update improves downstream value estimation. In particular, given $P_K \in \mathbb{R}^{d' \times 2d}$ that generates a perturbed input view and $P_V \in \mathbb{R}^{d' \times 2d}$ for the target, the self-supervised loss is defined as:

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} before the current test-time update. At each timestep t , VITA updates the parameters by

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . These sequential updates make f_{adapt} retain information from past observations in its parameters, providing temporal context without an explicit recurrent state. Section 4.6.2 confirms that these incremental parameter updates are substantially more effective than trajectory-level updates for value estimation.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

After test-time adaptation, a meta-learned linear projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps the input z_t into the adaptation space $\mathbb{R}^{d'}$. The adapted representation then passes through f_{adapt} and a regression head h , producing the goal-conditioned value estimate:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is a two-layer multilayer perceptron (MLP), matching value-function estimator architectures used in deep reinforcement learning (Espeholt et al., 2018). We train the network with normalized progress labels y_t from expert demonstrations as described in Section 2.1, using a supervised prediction loss ℓ_{pred} defined as the mean squared error between predicted values $V(z_t; g)$ and targets y_t . At inference, h remains frozen while f_{adapt} updates online under the self-supervised objective. This design gives the estimator zero-shot adaptability without requiring task-specific demonstrations.

3.2 TRAINING PROCEDURE

We train the value function estimator with gradient-based meta-learning so that the self-supervised loss ℓ_{self} produces test-time updates that reduce the supervised prediction loss ℓ_{pred} (Sun et al., 2024). Training uses diverse sub-trajectories selected by dissimilarity-based sampling, which directly counters shortcut learning and encourages the estimator to rely on semantic and temporal cues.

3.2.1 META-LEARNING

During training, the adaptation module f_{adapt} is updated online at each timestep using the self-supervised loss ℓ_{self} . Following gradient-based meta-learning (Finn et al., 2017), we differentiate

through these adaptation updates with respect to the supervised prediction loss ℓ_{pred} . This procedure jointly optimizes the initialization θ_0 of f_{adapt} , the linear projections P_K, P_V, P_Q , and the regression head h . The total training loss combines the supervised prediction loss ℓ_{pred} with the self-supervised loss ℓ_{self} , weighted by a scalar λ . The learned θ_0 becomes the inference-time initialization, so the estimator is explicitly trained to adapt its internal representation at test time and improve value estimation after adaptation.

3.2.2 DISSIMILARITY-BASED SAMPLING

Expert visual trajectories contain highly redundant consecutive frames, a well-known issue in video action recognition (Wang et al., 2018). This redundancy creates a concrete shortcut risk for value estimation (Geirhos et al., 2020; Lee et al., 2021b): the model can overfit to frequent late-stage visual patterns instead of learning robust semantic progress. We address this failure mode with dissimilarity-based sampling, which constructs mini-batches from the most visually diverse sub-trajectories within each trajectory. The strategy increases intra-batch variance and acts as an importance-sampling mechanism that deliberately favors semantic diversity.

Formally, given a sequence of multimodal representations $\{z_1, \dots, z_T\}$, we extract fixed-length sub-trajectories using a sliding window of size w_{tr} and stride s , yielding a candidate set $\mathcal{W}$. A diverse subset of size k maximizes pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i is a sub-trajectory of length w_{tr} . Solving Eq. 4 is intractable, requiring enumeration of all $\binom{|\mathcal{W}|}{k}$ subsets. To approximate this objective efficiently, we adopt a scoring-based heuristic, assigning each window $w \in \mathcal{W}$ a score equal to its total dissimilarity with all others:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

Thus, computing a diversity score s for each window and selecting the k highest-scoring windows gives a diverse subset with polynomial-time complexity. The overhead is negligible relative to model training, as shown in Appendix G. The ablation in Section 4.6.1 confirms that this sampling choice improves the ability to distinguish expert from non-expert visual trajectories.

4 EXPERIMENTS

We evaluate VITA on the capabilities that a useful goal-conditioned value function must satisfy: (i) robust value estimation for real-world robotic manipulation under task, environment, and embodiment shifts; (ii) reliable separation of expert and non-expert visual trajectories in real-world manipulation; and (iii) zero-shot reward shaping for offline RL on Meta-World MT10, a simulated benchmark for multi-task robot learning.

4.1 TRAINING SETUP

We train VITA on expert visual trajectories paired with natural language task descriptions from BridgeData V2 (Walke et al., 2023), which covers a broad set of manipulation tasks, environments, and robot embodiments. Our curated training subset contains 2,986 expert demonstrations for pick-and-place manipulation only; it deliberately excludes folding, sweeping, and stacking tasks that appear at evaluation time. All training demonstrations use a single robot embodiment (WidowX 250) across 4 ToyKitchen configurations. We reserve 287 additional expert demonstrations as the in-distribution test set, `tk_pnp`; Appendix A shows training examples. We use OpenCLIP ViT-B/32 as the frozen backbone for video-frame and task-description encoding. The training objective combines ℓ_{pred} and ℓ_{self} with $\lambda_{\text{self}} = 0.5$. During training, dissimilarity-based sampling uses window size $w_{\text{tr}} = 8$, $k = 8$ selected sub-trajectories, and stride $s = 1$. At test time, VITA uses one gradient step ($t_{\text{ep}} = 1$) with learning rate $\eta = 0.1$. Appendix D reports the full implementation details and hyperparameter sweeps.

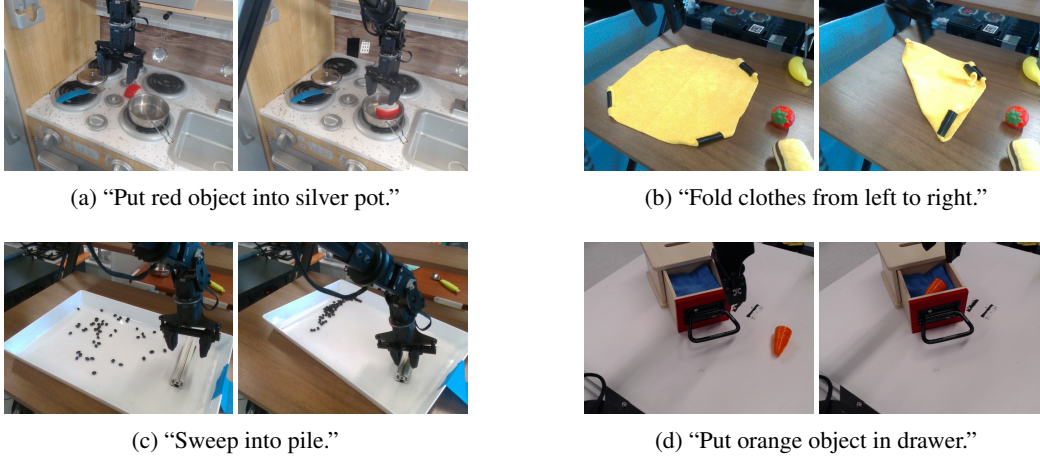


Figure 3: Representative visual trajectories and task descriptions used to test generalization across distribution shifts. (a) In-distribution. (b, c) Environment shift. (d) Embodiment and environment shift.

4.2 BASELINES

We compare VITA with value functions built from both contrastive and autoregressive VLMs. VLM-CL is the direct zero-shot CLIP similarity baseline, computing cosine similarity between each frame and the task description (Baumli et al., 2023). VLM-RM strengthens this similarity baseline by regularizing CLIP embeddings along the direction from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT trains a supervised regression head on frozen CLIP multimodal representations, and CLIP-GRU adds explicit recurrent temporal modeling with a Gated Recurrent Unit (GRU) (Cho et al., 2014). GVL (Ma et al., 2024) is the state-of-the-art zero-shot value function based on autoregressive VLMs. We evaluate GVL-0S in its zero-shot setting, where the prompt contains only the test trajectory and task description, and GVL-1S in its one-shot in-context setting, where a shuffled labeled trajectory from the training distribution is included as an example. Following GVL, we use Gemini 1.5 Pro (gemini-1.5 Pro-latest) (Gemini Team et al., 2024) with the proposed prompt template. We also tested Qwen-VL 2.5 (Yang et al., 2024), which did not overcome temporal bias, and GPT-4o (Hurst et al., 2024), which declined to produce scalar value estimates. Appendix D gives additional implementation details.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

Table 1: VOC scores show value-estimation generalization under distribution shifts. **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = Both Shifts.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
ES	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	0.658
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
EM	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	0.820
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
ES & EM	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

We test progress estimators on curated BridgeData V2 subsets that isolate generalization across environments, tasks, and robot embodiments. Environment shifts change scene layout or background: `lm_pnp` places pick-and-place in front of a laundry machine, `td_fold`, `ft_fold`, and `rd_fold` require cloth folding on new surfaces, and `ms_sweep` adds a long-horizon sweeping task in a confined tray. Embodiment shifts use the DeepThought robot, which differs from the training embodiment (WidowX 250) in morphology and camera perspective. `dt_tk_pnp` and `dt_tk_stack` preserve the ToyKitchen environment while changing embodiment, whereas `dt_ft_stack` and `dt_rd_pnp` combine embodiment and environment shifts. Each dataset has 200 expert trajectories with task descriptions, except `ms_sweep`, which has 100 because of dataset size. Appendix B details all evaluation datasets. We measure value-estimation quality with *Value Order Correlation* (VOC) (Ma et al., 2024), which quantifies alignment between predicted task progress and chronological frame order. Formally, let $p_1, \dots, p_T$ denote predicted progress values for a trajectory of T frames, and let $r_t = t$ represent the temporal index. VOC is the Spearman rank correlation ρ_s between predicted values and temporal indices.

Table 1 shows a clear pattern: methods that preserve temporal information, VITA and CLIP-GRU, consistently outperform the autoregressive VLM baselines GVL-0S and GVL-1S. This result identifies temporal order as a decisive ingredient for progress estimation and exposes the limitation of relying on prompted autoregressive VLMs alone. Among temporally aware methods, VITA outperforms CLIP-GRU on 6 of 10 datasets, showing that implicit memory through test-time adaptation is more robust than a supervised recurrent hidden state in most shifts. Both temporally aware methods outperform CLIP-based baselines without temporal modeling (CLIP-FT, VLM-CL, VLM-RM), confirming that semantic similarity alone is insufficient for value estimation. CLIP-FT is competitive with GVL in-distribution but does not transfer to out-of-distribution settings. In-context examples help GVL-0S in some cases but provide no reliable improvement and sometimes degrade performance. GVL performs well on folding tasks (`td_fold`, `ft_fold`, `rd_fold`) but falls behind on stacking (`dt_tk_stack`, `dt_ft_stack`) and pick-and-place (`dt_tk_pnp`, `lm_pnp`), suggesting a task-dependent bias in the autoregressive VLM. VITA, in contrast, maintains strong performance across task types and shifts, demonstrating stronger generalization than in-context learning in this evaluation.

The long-horizon `ms_sweep` task is difficult for nearly all methods, yet VITA achieves the best VOC score. Its advantage over CLIP-GRU on this task strengthens the conclusion that test-time adaptation is a more effective temporal-memory mechanism than recurrent hidden states for long-horizon manipulation in this setting. Under embodiment shifts, VITA also generalizes strongly. On `dt_tk_pnp`, which changes the robot while preserving environment and task family, both VITA and CLIP-GRU exceed their in-distribution performance, showing that learned temporal context can transfer across robot embodiments. By contrast, GVL-1S performs poorly on both `dt_tk_pnp` and `dt_tk_stack`, indicating that few-shot prompting is not sufficient for embodiment-shift generalization in our setup.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Beyond distribution-shift generalization, a value function must reject suboptimal behavior rather than merely track time. We therefore test whether VITA distinguishes expert from scripted non-expert visual trajectories by assigning lower progress-order scores to the latter. The scripted trajectories come from the same in-distribution BridgeData V2 setting as the expert data: they are collected in ToyKitchen with the WidowX 250 robot using a heavily randomized controller (Walke et al., 2023). They cover pick-and-place tasks with overlapping object categories, including general objects (`sc_pnp_obj`), rigid items (`sc_pnp_obj`), soft toys (`sc_pnp_stoy`), utensils (`sc_pnp_uten`), and a cloth (`sc_pnp_cloth`). Although these trajectories match the training distribution in task family, embodiment, and environment, their behavior lacks the smooth monotonic progress of expert demonstrations. Appendix C shows scripted examples. Because the scripted controller is randomized, the datasets do not encode calibrated suboptimality levels; nevertheless, a useful value estimator should assign lower VOC to these trajectories than to expert trajectories in the same setting. We measure discriminative success with *BinVOC*, defined as $1[\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$, which equals 1 when expert trajectories achieve higher VOC than suboptimal ones and 0 otherwise.

We report BINVOC averaged across scripted datasets. Expert performance (VOC_{exp}) is measured on `tk_pnp`, the in-distribution pick-and-place test set aligned with the scripted evaluation. VITA,

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-0S	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. Non-Expert (BinVOC).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on MT10 (IQM).

Table 2: (a) Discriminative success in separating expert from scripted robot demonstrations, measured by average BinVOC across 5 in-distribution scripted datasets. (b) Offline RL performance on Meta-World MT10, where VITA achieves the strongest interquartile mean (IQM) with 95% stratified bootstrap CIs.

GVL-0S, and GVL-1S achieve perfect discrimination, assigning higher VOC to expert trajectories for every scripted task. VITA also outperforms CLIP-GRU, supporting the claim that implicit memory from sequential test-time updates is less susceptible to temporal shortcuts than recurrent hidden-state memory in this discriminative test. The pure CLIP baselines (VLM-CL, VLM-RM) underperform because they do not model temporal structure, while CLIP-FT is stronger but still fails on one task, indicating sensitivity to object distribution shifts.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

We evaluate whether VITA can serve as a zero-shot dense reward for offline RL on Meta-World MT10 (Yu et al., 2020), which contains ten diverse manipulation tasks including pick-and-place, door opening, and pushing. This is a deliberately strong transfer test: VITA is trained on real-world robotic data and then used zero-shot in simulation. For each Meta-World MT10 task, we generate 20 expert visual demonstrations with Meta-World expert policies, and the goal-conditioned value estimator defines a dense reward for every visual observation. We train policies with Implicit Q-Learning (IQL) (Kostrikov et al., 2021) in the offline RL setting. Expert data generation and policy training are repeated across 10 random seeds. After training, each multi-task policy is evaluated on 20 rollouts per task, and we compute the average return for each (task, seed) pair. We report the interquartile mean (IQM) across all pairs with 95% stratified bootstrap confidence intervals, following Agarwal et al. (2021). Appendix E provides the training details and evaluation protocol.

Table 2 shows that VITA achieves the highest MT10 IQM return (0.815), outperforming every CLIP-based baseline. We do not compare directly with GVL at RL scale because Gemini-1.5 is a proprietary autoregressive VLM with prohibitive inference cost in this setting, and the open-source alternatives we tested did not produce reliable progress estimates. We also include the Meta-World fuzzy-logic dense reward (META-WL), which reaches 0.779. VITA surpasses this simulator-provided reward, demonstrating that a value estimator trained on real-world videos can transfer into simulated reward shaping strongly enough to improve downstream policy learning.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We validate dissimilarity-based sampling by isolating how training-time sub-trajectory selection affects VITA’s ability to distinguish expert from non-expert demonstrations. Full-trajectory sampling (Sun et al., 2024) computes the adaptation loss over all overlapping sub-trajectories with stride $s = 1$; in our setup, this encourages overfitting to global temporal shortcuts. Random sampling increases diversity by selecting sub-trajectories uniformly, but it does not enforce semantic diversity. Dissimilarity-based sampling explicitly selects semantically diverse sub-trajectories and therefore generalizes better. The proposed pairwise-dissimilarity criterion outperforms both full-trajectory and random sampling for expert-versus-non-expert discrimination. Appendix F.1 gives the full ablation.

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We analyze temporal memory by comparing test-time adaptation strategies that preserve, reset, or average temporal context. VITA performs step-by-step adaptation without reset (implicit memory), incrementally updating parameters from θ_{t-1} so the value estimator stores trajectory history directly in its parameters and preserves both temporal order and accumulated context. We compare it with three alternatives: (i) *trajectory-level adaptation (TTT-TR)*, which applies one gradient update over the full trajectory and therefore averages the loss across frames, discarding temporal order; (ii) *memoryless adaptation (TTT-RS)*, which resets the adaptation module to θ_0 at every step and adapts only to the current observation, preventing temporal carryover; (iii) *explicit memory (TTT-EX)*, which resets at every step and updates on a recent-frame window matching the meta-training window size ($w_{tr} = 8$). Across all datasets, VITA consistently outperforms trajectory-level, memoryless, and explicit-memory variants, establishing sequential implicit memory as the strongest temporal mechanism for value estimation in our experiments. Appendix F.2 provides the full analysis.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

Contrastive VLMs have already been used as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024), but they rely on frame-level similarity and therefore lack temporal modeling. Domain-specific CLIP fine-tuning (Fan et al., 2022; Jiang et al., 2024) can produce dense rewards, but the required supervision weakens zero-shot applicability. RoboCLIP (Sontakke et al., 2023) supports one-shot reward specification without fine-tuning, yet it yields sparse rewards and still requires demonstrations. Large-scale multimodal pre-training methods (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) seek generalization through scale, while ReWiND (Zhang et al., 2025b) trains a cross-modal sequential transformer on augmented large-scale data with video rewinding and uses a few in-domain demonstrations to improve transfer. VITA takes a different position: it adapts a pre-trained contrastive vision–language encoder at inference time, capturing semantic and temporal context without additional large-scale multimodal pre-training or domain-specific demonstrations. Because the adaptation mechanism is encoder-agnostic, VITA can strengthen any pre-trained vision–language representation that supports value estimation. Autoregressive VLMs (Alayrac et al., 2022) use in-context learning over trajectories and have been applied to success detection (Du et al., 2023) and goal-conditioned value functions (Ma et al., 2024). GVL (Ma et al., 2024) reduces monotonic bias by shuffling frames at inference, but this remedy discards the temporal order needed for temporal reasoning and for distinguishing visually similar but temporally distinct states. VITA preserves chronological order and captures context through test-time adaptation instead of removing the temporal signal.

5.2 TEST-TIME ADAPTATION

Test-time adaptation provides a parameter-efficient route for adapting VLMs without full fine-tuning. Prompt-based methods optimize parameters at test time from task context, but their mechanisms remain difficult to interpret (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Lim et al. (2025) argue that CLIP representations are suitable for test-time adaptation because CLIP encodes human-interpretable concepts identified through mechanistic interpretability (Bricken et al., 2023). CLIP-based similarity has also served as a reward signal whose parameters are optimized at test time with reinforcement learning (Zhao et al., 2024). Test-time training (Sun et al., 2020) offers the mechanism we need: an adaptation module updated at each timestep by a self-supervised loss. Unlike meta-reinforcement learning methods (Finn et al., 2017; Bauer et al., 2023), which require task-specific adaptation episodes during training, test-time training enables direct online adaptation during inference without task labels.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

Our results establish test-time adaptation as an effective mechanism for zero-shot value functions in robotic manipulation. VITA generalizes across task, environment, and embodiment shifts, distinguishes expert from non-expert visual trajectories, and supplies zero-shot rewards that improve offline RL. Sequential updates encode history into the adaptation parameters, allowing VITA to capture temporal and semantic context more effectively than pure CLIP baselines and prompted autoregressive VLMs. Its advantages over CLIP-GRU on expert–non-expert discrimination and offline RL further indicate that implicit memory from sequential test-time updates generalizes better than recurrent hidden-state memory for zero-shot value estimation. Pre-trained autoregressive VLMs such as GVL retain broad semantic knowledge, but they are not explicitly trained for progress estimation over visual trajectories. In our setup, the per-timestep adaptation cost remains negligible because only a lightweight module is updated, preserving real-time feasibility.

6.2 LIMITATIONS AND FUTURE WORK

The evidence supports test-time adaptation as a strong route to zero-shot progress estimation, but the method is not without limits. Settings with high execution variability or substantially longer horizons may still challenge sequential adaptation. Although the lightweight adapter makes overhead negligible in our experiments, updating a value estimator at every timestep can raise deployment-safety concerns in real-time robotic systems. Future work should explore alternative adaptation rules for vision–language models, especially for agents trained inside world models, and should test VITA in closed-loop control and more complex RL environments. Because VITA is agnostic to the underlying vision–language encoder, the same meta-learned adaptation principle could strengthen other domain-specific multimodal representations. We also empirically show that dissimilarity-based sampling mitigates shortcut learning, but a theoretical account of why diversity-based sampling reduces shortcuts remains open. That analysis is separate from our central contribution: a value function estimator that meta-learns test-time adaptation to improve zero-shot generalization and temporal reasoning.

REFERENCES

- Niket Agarwal, Arslan Ali, Maciej Bala, Yogesh Balaji, Erik Barker, Tiffany Cai, Prithvijit Chatopadhyay, Yongxin Chen, Yin Cui, Yifan Ding, et al. Cosmos world foundation model platform for physical ai. *arXiv preprint arXiv:2501.03575*, 2025.
- Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron C Courville, and Marc Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems*, 2021.
- Minttu Alakuijala, Reginald McLean, Isaac Woungang, Nariman Farsad, Samuel Kaski, Pekka Marttinen, and Kai Yuan. Video-language critic: Transferable reward functions for language-conditioned robotics. *arXiv preprint arXiv:2405.19988*, 2024.
- Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andy Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. Flamingo: a visual language model for few-shot learning. *arXiv preprint arXiv:2204.14198*, 2022.
- Bowen Baker, Ilge Akkaya, Peter Zhokov, Joost Huizinga, Jie Tang, Adrien Ecoffet, Brandon Houghton, Raul Sampedro, and Jeff Clune. Video pretraining (vpt): Learning to act by watching unlabeled online videos. In *Advances in Neural Information Processing Systems*, 2022.
- Jakob Bauer, Kate Baumli, Feryal M. P. Behbahani, Avishkar Bhoopchand, Nathalie Bradley-Schmieg, Michael Chang, Natalie Clay, Adrian Collister, Vibhavari Dasagi, Lucy Gonzalez, Karol Gregor, Edward Hughes, Sheleem Kashem, Maria Loks-Thompson, Hannah Openshaw, Jack Parker-Holder,

Shreya Pathak, Nicolas Perez Nieves, Nemanja Rakicevic, Tim Rocktäschel, Yannick Schroecker, Satinder Singh, Jakub Sygnowski, Karl Tuyls, Sarah York, Alexander Zacherl, and Lei M. Zhang. Human-timescale adaptation in an open-ended task space. In *International Conference on Machine Learning*, 2023.

Kate Baumli, Satinder Baveja, Feryal M. P. Behbahani, Harris Chan, Gheorghe Comanici, Sebastian Flennerhag, Maxime Gazeau, Kristian Holsheimer, Dan Horgan, Michael Laskin, Clare Lyle, Hussain Masoom, Kay McKinney, Volodymyr Mnih, Alexander Neitz, Fabio Pardo, Jack Parker-Holder, John Quan, Tim Rocktäschel, Himanshu Sahni, Tom Schaul, Yannick Schroecker, Stephen Spencer, Richie Steigerwald, Luyu Wang, and Lei Zhang. Vision-language models as a source of rewards. *arXiv preprint arXiv:2312.09187*, 2023.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermy, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E. Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Krzysztof Choromanski, Tianli Ding, Danny Driess, Kumar Avinava Dubey, Chelsea Finn, Peter R. Florence, Chuyuan Fu, Montse Gonzalez Arenas, Keerthana Gopalakrishnan, Kehang Han, Karol Hausman, Alexander Herzog, Jasmine Hsu, Brian Ichter, Alex Irpan, Nikhil J. Joshi, Ryan C. Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Sergey Levine, Henryk Michalewski, Igor Mordatch, Karl Pertsch, Kanishka Rao, Krista Reymann, Michael S. Ryoo, Grecia Salazar, Pannag R. Sanketi, Pierre Sermanet, Jaspier Singh, Anikait Singh, Radu Soricut, Huong Tran, Vincent Vanhoucke, Quan Ho Vuong, Ayzaan Wahid, Stefan Welker, Paul Wohlhart, Ted Xiao, Tianhe Yu, and Brianna Zitkovich. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning*, 2023.

Jake Bruce, Michael D Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, Yusuf Aytar, Sarah Maria Elisabeth Bechtle, Feryal Behbahani, Stephanie C.Y. Chan, Nicolas Heess, Lucy Gonzalez, Simon Osindero, Sherjil Ozair, Scott Reed, Jingwei Zhang, Konrad Zolna, Jeff Clune, Nando De Freitas, Satinder Singh, and Tim Rocktäschel. Genie: Generative interactive environments. In *International Conference on Machine Learning*, 2024.

Annie S Chen, Suraj Nair, and Chelsea Finn. Learning generalizable robotic reward functions from in-the-wild human videos. *arXiv preprint arXiv:2103.16817*, 2021.

Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using rnn encoder-decoder for statistical machine translation. *arXiv preprint arXiv:1406.1078*, 2014.

Nitish Dashora, Dibya Ghosh, and Sergey Levine. Viva: Video-trained value functions for guiding online rl from diverse data. *arXiv preprint arXiv:2503.18210*, 2025.

Yuqing Du, Ksenia Konyushkova, Misha Denil, Akhil Raju, Jessica Landon, Felix Hill, Nando de Freitas, and Serkan Cabi. Vision-language models as success detectors. In *Conference on Lifelong Learning Agents*, 2023.

Ashley Edwards, Himanshu Sahni, Yannick Schroecker, and Charles Isbell. Imitating latent policies from observation. In *International Conference on Machine Learning*, 2019.

Lasse Espeholt, Hubert Soyer, Remi Munos, Karen Simonyan, Vlad Mnih, Tom Ward, Yotam Doron, Vlad Firoiu, Tim Harley, Iain Dunning, Shane Legg, and Koray Kavukcuoglu. IMPALA: Scalable distributed deep-RL with importance weighted actor-learner architectures. In *International Conference on Machine Learning*, 2018.

Linxi Fan, Guanzhi Wang, Yunfan Jiang, Ajay Mandlekar, Yuncong Yang, Haoyi Zhu, Andrew Tang, De-An Huang, Yuke Zhu, and Anima Anandkumar. Minedojo: Building open-ended embodied agents with internet-scale knowledge. In *Advances in Neural Information Processing Systems*, 2022.

-
- Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *International Conference on Machine Learning*, 2017.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2020.
- Gemini Team, Petko Georgiev, Ving Ian Lei, Ryan Burnell, Libin Bai, Anmol Gulati, Garrett Tanzer, Damien Vincent, Zhufeng Pan, Shibo Wang, et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.
- Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep learning*. MIT Press, 2016.
- David Ha and Jürgen Schmidhuber. World models. In *Advances in Neural Information Processing Systems*, 2018.
- Danijar Hafner, Timothy Lillicrap, Mohammad Norouzi, and Jimmy Ba. Mastering atari with discrete world models. In *International Conference on Learning Representations*, 2021.
- Jonathan Ho and Stefano Ermon. Generative adversarial imitation learning. In *Advances in Neural Information Processing Systems*, 2016.
- Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural computation*, 1997.
- Anthony Hu, Lloyd Russell, Hudson Yeo, Zak Murez, George Fedoseev, Alex Kendall, Jamie Shotton, and Gianluca Corrado. GAIA-1: A generative world model for autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- Edward Hughes, Michael D. Dennis, Jack Parker-Holder, Feryal M. P. Behbahani, Aditi Mavalankar, Yuge Shi, Tom Schaul, and Tim Rocktäschel. Position: Open-endedness is essential for artificial superhuman intelligence. In *International Conference on Machine Learning*, 2024.
- Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*, 2024.
- Haobin Jiang, Junpeng Yue, Hao Luo, Ziluo Ding, and Zongqing Lu. Reinforcement learning friendly vision-language model for minecraft. In *European Conference on Computer Vision*, 2024.
- Muhammad Uzair Khattak, Hanoona Rasheed, Muhammad Maaz, Salman Khan, and Fahad Shahbaz Khan. Maple: Multi-modal prompt learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Paul Foster, Pannag R. Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. Openvla: An open-source vision-language-action model. In *Conference on Robot Learning*, 2024.
- Ilya Kostrikov, Ashvin Nair, and Sergey Levine. Offline reinforcement learning with implicit q-learning. *arXiv preprint arXiv:2110.06169*, 2021.
- Youngwoon Lee, Andrew Szot, Shao-Hua Sun, and Joseph J. Lim. Generalizable imitation learning from observation via inferring goal proximity. In *Advances in Neural Information Processing Systems*, 2021a.
- Youngwoon Lee, Andrew Szot, Shao-Hua Sun, and Joseph J. Lim. Generalizable imitation learning from observation via inferring goal proximity. In *Advances in Neural Information Processing Systems*, 2021b.
- Sergey Levine, Aviral Kumar, George Tucker, and Justin Fu. Offline reinforcement learning: Tutorial, review, and perspectives on open problems. *arXiv preprint arXiv:2005.01643*, 2020.

648 Jianxiong Li, Jinliang Zheng, Yinan Zheng, Liyuan Mao, Xiao Hu, Si Cheng, Haoyi Niu, Jihao Liu,
649 Yu Liu, Jingjing Liu, Ya-Qin Zhang, and Xianyu Zhan. Decisionncc: Embodied multimodal
650 representations via implicit preference learning. In *International Conference on Machine Learning*,
651 2024.

652 Hyesu Lim, Jinho Choi, Jaegul Choo, and Steffen Schneider. Sparse autoencoders reveal selec-
653 tive remapping of visual concepts during adaptation. In *International Conference on Learning*
654 *Representations*, 2025.

656 Ziyi Lin, Honglie Liu, Andrew Zisserman, and Andrea Vedaldi. Frozen clip models are efficient
657 video learners. In *European Conference on Computer Vision*, 2022.

659 Xudong Ma, Chen Liang, Zhecan Wang, Chuang Gan, and Zhiting Li. Swapprompt: Test-time
660 prompt adaptation for vision-language models. In *Advances in Neural Information Processing*
661 *Systems*, 2023a.

662 Yecheng Jason Ma, Shagun Sodhani, Dinesh Jayaraman, Osbert Bastani, Vikash Kumar, and Amy
663 Zhang. Vip: Towards universal visual reward and representation via value-implicit pre-training.
664 *arXiv preprint arXiv:2210.00030*, 2022.

666 Yecheng Jason Ma, Vikash Kumar, Amy Zhang, Osbert Bastani, and Dinesh Jayaraman. LIV:
667 Language-image representations and rewards for robotic control. In *International Conference on*
668 *Machine Learning*, 2023b.

669 Yecheng Jason Ma, Joey Hejna, Ayzaan Wahid, Chuyuan Fu, Dhruv Shah, Jacky Liang, Zhuo Xu,
670 Sean Kirmani, Peng Xu, Danny Driess, Ted Xiao, Jonathan Tompson, Osbert Bastani, Dinesh
671 Jayaraman, Wenhao Yu, Tingnan Zhang, Dorsa Sadigh, and Fei Xia. Vision language models are
672 in-context value learners. In *International Conference on Learning Representations*, 2024.

674 Yutaka Matsuo, Yann LeCun, Maneesh Sahani, Doina Precup, David Silver, Masashi Sugiyama, Eiji
675 Uchibe, and Jun Morimoto. Deep learning, reinforcement learning, and world models. *Neural*
676 *Networks*, 2022.

677 Viorica Pătrăucean, Yusuf Aytar, Roozbeh Mottaghi, Oleh Rybkin, Mario Lucic, Carl Vondrick,
678 William T. Freeman, and Andrew Zisserman. Perception test: A diagnostic benchmark for
679 multimodal video models. In *Advances in Neural Information Processing Systems*, 2023.

681 Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal,
682 Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever.
683 Learning transferable visual models from natural language supervision. In *International Conference*
684 *on Machine Learning*, 2021.

685 Scott Reed, Konrad Zolna, Emilio Parisotto, Sergio Gomez Colmenarejo, Alexander Novikov, Gabriel
686 Barth-Maron, Mai Giménez, Yury Sulsky, Jackie Kay, Jost Tobias Springenberg, Tom Eccles,
687 Jake Bruce, Ali Razavi, Ashley D. Edwards, Nicolas Manfred Otto Heess, Yutian Chen, Raia
688 Hadsell, Oriol Vinyals, Mahyar Bordbar, and Nando de Freitas. A generalist agent. *arXiv preprint*
689 *arXiv:2205.06175*, 2022.

691 Juan Rocamonde, Victoriano Montesinos, Elvis Nava, Ethan Perez, and David Lindner. Vision-
692 language models are zero-shot reward models for reinforcement learning. In *International Confer-*
693 *ence on Learning Representations*, 2024.

694 Dominik Schmidt and Minqi Jiang. Learning to act without actions. In *International Conference on*
695 *Learning Representations*, 2024.

697 David Silver and Richard S. Sutton. Welcome to the era of experience. Technical report, Google AI,
698 2025.

700 SIMA Team, Maria Abi Raad, Arun Ahuja, Catarina Barros, Frederic Besse, Andrew Bolt, Adrian
701 Bolton, Bethanie Brownfield, Gavin Buttimore, Max Cant, et al. Scaling instructable agents across
many simulated worlds. *arXiv preprint arXiv:2404.10179*, 2024.

-
- Sumedh Sontakke, Jesse Zhang, Sébastien M. R. Arnold, Karl Pertsch, Erdem Biyik, Dorsa Sadigh, Chelsea Finn, and Laurent Itti. Roboclip: One demonstration is enough to learn robot policies. In *Advances in Neural Information Processing Systems*, 2023.
- Yu Sun, Xiaolong Wang, Zhuang Liu, John Miller, Alexei A. Efros, and Moritz Hardt. Test-time training with self-supervision for generalization under distribution shifts. In *International Conference on Machine Learning*, Proceedings of Machine Learning Research. PMLR, 2020.
- Yu Sun, Xinhao Li, Karan Dalal, Jiarui Xu, Arjun Vikram, Genghan Zhang, Yann Dubois, Xinlei Chen, Xiaolong Wang, Sanmi Koyejo, Tatsunori Hashimoto, and Carlos Guestrin. Learning to (learn at test time): Rnns with expressive hidden states. *arXiv preprint arXiv:2407.04620*, 2024.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.
- Homer Rich Walke, Kevin Black, Tony Z. Zhao, Quan Vuong, Chongyi Zheng, Philippe Hansen-Estruch, Andre Wang He, Vivek Myers, Moo Jin Kim, Max Du, Abraham Lee, Kuan Fang, Chelsea Finn, and Sergey Levine. Bridgedata v2: A dataset for robot learning at scale. In *Conference on Robot Learning*, 2023.
- Limin Wang, Yuanjun Xiong, Dahua Lin, and Luc Van Gool. Temporal segment networks: Towards good practices for deep action recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2018.
- Renhao Wang, Yu Sun, Yossi Gandelsman, Xinlei Chen, Alexei A. Efros, and Xiaolong Wang. Test-time training on video streams. *arXiv preprint arXiv:2307.05014*, 2023.
- An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- Ze Yang, Yiming Zhang, Jerry Yu, Junru Cai, and Ming-Hsuan Yang. UniSim: A neural closed-loop sensor simulator. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- Tianhe Yu, Deirdre Quillen, Zhanpeng He, Ryan Julian, Karol Hausman, Chelsea Finn, and Sergey Levine. Meta-world: A benchmark and evaluation for multi-task and meta reinforcement learning. In *Conference on Robot Learning*, 2020.
- Chuheng Zhang, Tim Pearce, Pushi Zhang, Kaixin Wang, Xiaoyu Chen, Wei Shen, Li Zhao, and Jiang Bian. What do latent action models actually learn? *arXiv preprint arXiv:2506.15691*, 2025a.
- Jiahui Zhang, Yusen Luo, Abrar Anwar, Sumedh Anand Sontakke, Joseph J Lim, Jesse Thomason, Erdem Biyik, and Jesse Zhang. RewiND: Language-guided rewards teach robot policies without new demonstrations. In *Conference on Robot Learning*, 2025b.
- Shuyuan Zhao, Xin Wang, Linchao Zhu, and Yi Yang. Test-time adaptation with clip reward for zero-shot generalization in vision-language models. *arXiv preprint arXiv:2401.12345*, 2024.
- Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu. Learning to prompt for vision-language models. *International Journal of Computer Vision*, 2022.

A TRAINING DATASET

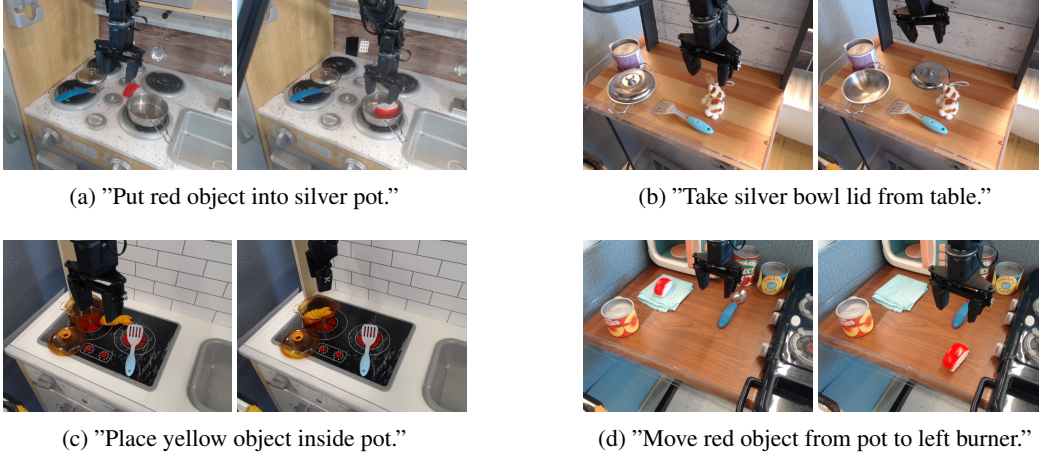


Figure 4: Training examples show start and end frames from expert demonstrations with their natural language task descriptions, spanning four ToyKitchen configurations used to train VITA.

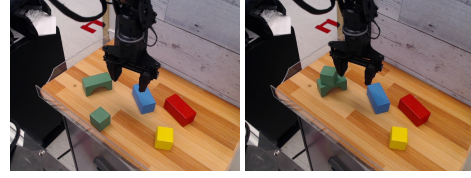
B EVALUATION DATASETS

Table 3: Evaluation datasets isolate task, environment, and embodiment shifts.

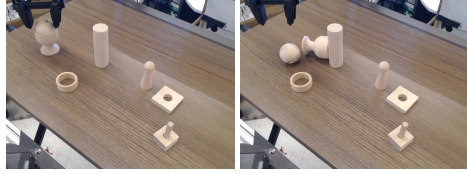
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



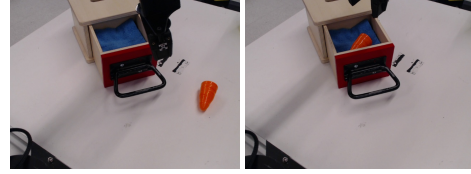
(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."

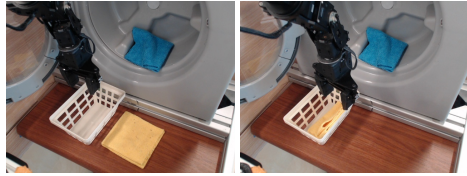


(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

Figure 5: Embodiment-shift examples show start and end frames with task descriptions. The top row changes the robot while preserving ToyKitchen, and the bottom row combines the DeepThought embodiment with new environments.



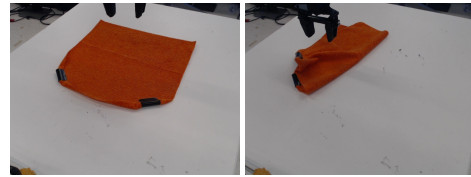
(a) lm_pnp: "Place blue cloth inside washer."



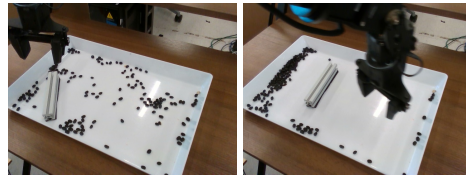
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



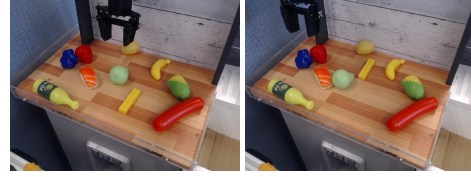
(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Environment-shift examples show start and end frames with task descriptions, holding the robot embodiment fixed while varying visual and structural scene context.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Scripted evaluation examples show start and end frames collected in ToyKitchen with the WidowX 250 robot. These pick-and-place trajectories vary object category, and generic task descriptions are used because public annotations are unavailable.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

We use the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset as a frozen backbone. The CLIP vision and text encoders produce representations for each visual observation and task description, respectively. A joint concatenation-based representation outperformed an element-wise product on validation, so we use concatenation throughout. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activation and projection dimension $d' = 64$. We train with AdamW using learning rate 1×10^{-4} , weight decay 1×10^{-4} , and a cosine learning-rate schedule with 10% warmup. The batch size is 32, and trajectories are padded to 120 frames, matching the longest training sequence. The self-supervised loss weight λ_{self} is set to 0.5 from validation performance. Training runs for 5 epochs because validation VOC plateaus early and longer training does not improve performance. Unless otherwise specified, dissimilarity-based sampling uses stride $s = 1$, sub-trajectory length $w_{\text{tr}} = 8$, and $k = 8$ selected windows. All experiments ran on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference time, VITA adapts only the temporal adaptation module f_{adapt} , keeping the training projection dimension ($d' = 64$). We use a single gradient step ($t_{\text{ep}} = 1$) with learning rate $\eta = 0.1$. This choice comes from a sweep over learning rates $\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$ and adaptation steps $\{1, 5, 10\}$, with validation results in Table 4. The sweep shows two competitive regimes: one larger step ($\eta = 0.1$, $t_{\text{ep}} = 1$) and multiple smaller steps ($\eta = 0.01$, $t_{\text{ep}} = 10$). We select the single-step setting because it is faster and achieves slightly stronger validation performance.

Table 4: Validation VOC identifies the strongest test-time learning rate η and number of adaptation steps t_{ep} for VITA.

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

D.3 BASELINES

VITA uses $t_{\text{ep}} = 1$, projection dimension $d' = 64$, and learning rate η of 0.1, selected by the validation sweep above. CLIP-FT keeps VITA’s frozen CLIP encoder and progress regressor structure but removes the adaptation module and self-supervised loss. It uses a frozen CLIP encoder followed by a linear projection and two-layer MLP trained with standard supervised regression, without meta-learning or test-time adaptation. To give this baseline substantial capacity, it uses an $8\times$ larger projection matrix and $10\times$ more training steps than our method (Lin et al., 2022). For VLM-RM, we use a baseline prompt describing the environment. CLIP-GRU also follows VITA’s frozen CLIP encoder and regression head h , but replaces f_{adapt} with a single-layer GRU of hidden size 64, explicitly storing temporal history in recurrent hidden states before passing the final hidden state to h . For both GVL-0S and GVL-1S, we use the latest Gemini 1.5 Pro, gemini-1.5-pro-latest (Gemini Team et al., 2024); the original GVL implementation used gemini-1.5-pro. We also evaluated GVL with Qwen-VL 2.5 (Yang et al., 2024), which did not overcome temporal bias despite frame shuffling, while GPT-4o (Hurst et al., 2024) consistently declined to produce scalar progress estimates in our setup. All evaluations used the prompt template from the original GVL work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

We use Implicit Q-Learning (IQL) with a fixed configuration across all tasks. Expectile regression uses $\tau = 0.7$, advantage weighting temperature is 3.0, and the clipping threshold is 100.0. Actor and critic learning rates are 1×10^{-4} and 3×10^{-4} , respectively, with batch size 256. Policies train for 100,000 gradient steps and are evaluated every 10,000 steps on 20 rollouts per task with horizon 150. Implementation follows d3rlpy, with configuration: expectile=0.7, weight_temp=3.0, max_weight=100.0, actor_lr=1e-4, critic_lr=3e-4, batch_size=256.

E.2 EVALUATION PROTOCOL

Following Agarwal et al. (Agarwal et al., 2021), we report the interquartile mean (IQM) of returns across all task–seed pairs. IQM averages the middle 50% of scores and is therefore less sensitive to outliers than the mean or median. We estimate 95% confidence intervals with 10,000 stratified bootstrap resamples, stratified by task. The same protocol is applied to every method in the Meta-World MT10 experiments.

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

Table 5 validates dissimilarity-based sampling by isolating how training-time sub-trajectory selection affects VITA’s ability to distinguish expert from non-expert demonstrations. Full-trajectory sampling (Sun et al., 2024) computes the adaptation loss over all overlapping sub-trajectories with

Table 5: Disc@VOC confirms how training-time sampling controls discriminative test-time adaptation. A checkmark denotes successful expert-versus-suboptimal discrimination; **FT**, **Rand**, and **Diss** denote full-trajectory, random, and dissimilarity-based sampling parameterized by (w_{tr}, k) .

Dataset	FT (8,-)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc_pnp_obj		✓	✓		✓
sc_pnp_robj	✓	✓	✓	✓	✓
sc_pnp_stoy			✓		✓
sc_pnp_uten	✓	✓	✓	✓	✓
sc_pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

stride $s = 1$; in our setup, this encourages overfitting to global temporal shortcuts. Random sampling increases diversity by selecting sub-trajectories uniformly, but it does not enforce semantic diversity. Dissimilarity-based sampling explicitly selects semantically diverse sub-trajectories and therefore generalizes better. The proposed pairwise-dissimilarity criterion outperforms both full-trajectory and random sampling for expert-versus-non-expert discrimination.

The results show that dissimilarity-based sampling with $w_{tr} = 8$ and $k = 8$ achieves perfect discriminative performance. Using fewer windows ($k = 4$) at the same length preserves this performance, whereas shortening the sub-trajectory to $w_{tr} = 4$ reduces effectiveness. Full-trajectory sampling performs worst, confirming that adapting over all consecutive windows encourages overfitting to global temporal cues. Random sampling improves over full-trajectory sampling but remains less reliable than dissimilarity-based selection. These results make sub-trajectory diversity and semantic locality central to discriminative test-time adaptation.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

Table 6: Validation VOC scores show that sequential implicit memory is the decisive adaptation strategy under distribution shifts. ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = Environment and Embodiment Shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

We compare four adaptation strategies: (1) TTT-TR, which performs one full-trajectory gradient update (offline) and averages the loss across frames, discarding temporal order; (2) TTT-RS, which resets the adapter at every step and updates only on the current observation, making it memoryless; (3) TTT-EX, which resets at every step and updates on a recent-frame window matching meta-training ($w_{tr} = 8$), thereby averaging gradients and losing order within the window; (4) VITA (OURS), which updates incrementally at every timestep and stores temporal history implicitly in adapter parameters. We apply the same test-time hyperparameter sweeps to all variants, as described in Appendix D. VITA, TTT-RS, and TTT-TR perform best with learning rate 0.1 and one adaptation step, whereas TTT-EX performs best with learning rate 1.0 and one adaptation step.

Table 6 makes the temporal-memory result unambiguous: TTT-TR, TTT-RS, and TTT-EX perform similarly to one another, while VITA consistently outperforms all three. Batched updates over either a full trajectory or a short explicit window behave much like the memoryless baseline, whereas sequential per-timestep updates provide the temporal accumulation required for accurate value estimation. For implicit memory, updating on a local window would double-count recent observations, once through the explicit window and once through accumulated parameters; updating less frequently would omit intermediate frames. The implicit-memory paradigm therefore relies on sequential, per-timestep updates without reset.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Let $\mathcal{W}$ denote the candidate set with size $N_c = |\mathcal{W}|$. To approximate diverse subset selection without the exponential cost of enumerating all $\binom{N_c}{b}$ subsets in Eq. 4, we use a scoring heuristic that assigns each candidate window a diversity score equal to the sum of its pairwise distances to all other windows, i.e., the row sum of the distance matrix. With $N_c = T - w_{\text{tr}} + 1$ candidate windows, each flattened window has dimension $m = w_{\text{tr}} \cdot d$, where d is the dimension of each multimodal representation z_t . We compute the full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ on the GPU using batched `cdist`, requiring $\mathcal{O}(N_c^2 \cdot w_{\text{tr}} \cdot d)$ operations.

In our experiments, training trajectories are short (mean length 69, max length 120), so the resulting overhead is small. With batch size 32, feature dimension $d = 1024$ for the CLIP-based multimodal representation, and window length $w_{\text{tr}} = 8$, the worst-case computational cost is:

$$32 \cdot N_c^2 \cdot w_{\text{tr}} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

This cost is negligible compared with a single forward and backward pass through the model, while it directly enables diverse-window selection.